AF-400 INVERTER DRIVES

INVERTER ASSEMBLY - 331X403AAG01 (100 KVA)
G02 (50 KVA)

OPERATION — TROUBLE SHOOTING — REPAIR

GEK-62517

DIAGRAMS

INVERTER ASSEMBLY	ELEM	36C764880AA
	CONN	36D868887AA, Sh.2
PHASE MODULE	ELEM	36C764878AA
	CONN	36C764884AA
CONVERTER MODULE	ELEM	36C764879AA
	CONN	36C764889AA
COMM. POWER SUPPLY	ELEM	36C764881AA
	CONN	36D868891AA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

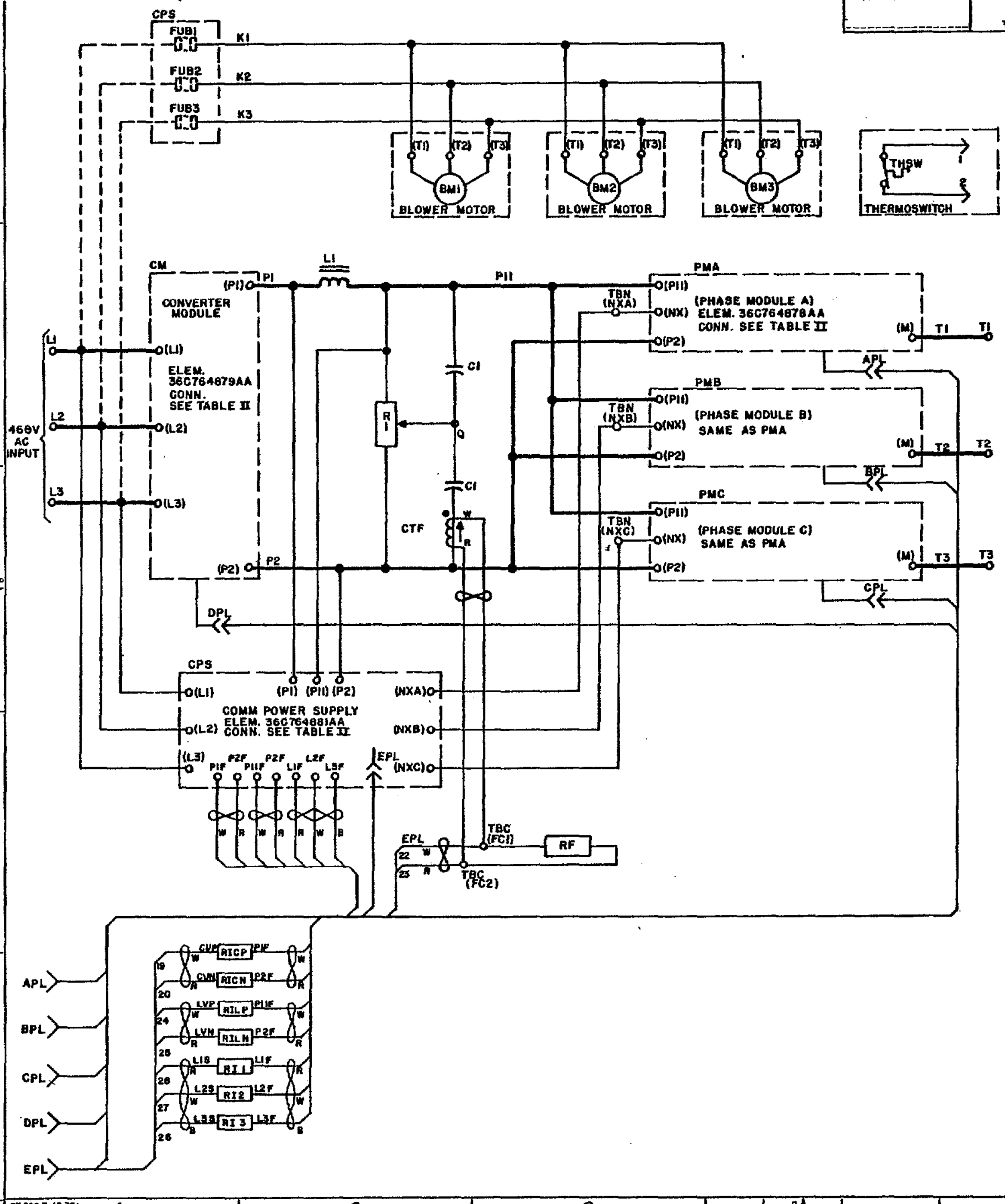
demark of the General Electric Co.

GENERAL ELECTRIC

36C764880AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 INVERTER ASMI
FIRST MADE FOR AF-400
331X401, 402, 403AA



CPS
FUB1
FUB2
FUB3
K1
K2
K3
(T1)
(T2)
(T3)
BM1
BM2
BM3
BLOWER MOTOR
BLOWER MOTOR
BLOWER MOTOR
THSW
THERMOSWITCH
OVER TEMP. IND.
CM
(P1)
P1
CONVERTER MODULE
(L1)
ELEM. 36C764879AA CONN. SEE TABLE II
(L2)
(L3)
(P2)
P2
L1
L2
L3
460V AC INPUT
L1
P11
C1
C1
R1
Q
CTF
W
R
PMA
(P11)
(NX)
(P2)
(PHASE MODULE A) ELEM. 36C764878AA CONN. SEE TABLE II
TBN (NXA)
(M)
T1
T1
APL
PMB
(PHASE MODULE B) SAME AS PMA
TBN (NXB)
T2
T2
BPL
PMC
(PHASE MODULE C) SAME AS PMA
TBN (NXC)
T3
T3
CPL
DPL
CPS
(L1)
(L2)
(L3)
(P1)
(P11)
(P2)
(NXA)
(NXB)
(NXC)
COMM POWER SUPPLY ELEM. 36C764881AA CONN. SEE TABLE II
P1F
P2F
P11F
P2F
L1F
L2F
L3F
EPL
W
R
B
EPL
22
23
TBC (FC1)
TBC (FC2)
RF
APL
BPL
CPL
DPL
EPL
19
20
24
25
26
27
28
CVP
RICP
P1F
CVN
RICN
P2F
LVP
RILP
P11F
LVN
RILN
P2F
L1S
RI1
L1F
L2S
RI2
L2F
L3S
RI3
L3F

NOTES:

1. THE ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON KVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.
2. THE L1 CHOKE IS MOUNTED REMOTE TO THE INVERTER ASSEMBLY. ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.
3. ELEMENTARY WIRING SYMBOLS
   - W R —INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER.
   - 1 2 3 —INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - →≻— —QUICK DISCONNECT "STAB-ON" CONNECTION.
   - —DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5. RESISTORS RIC, RIL, RI1, RI2, RI3 ARE, RESISTANCE ISOLATOR RESISTORS, AND ARE MOUNTED AS AN INTEGRAL PART OF THE WIRE HARNESS.
6. BM1, BM2, BM3 USED ON 300/400 KVA
   BM1, BM2 USED ON 200 KVA
   BM1 USED ON 100/50KVA
7. SEE TABLE II FOR CONNECTION DIAGRAMS.
8. TABLE I

KVA	331X	L1	C1//C1	C1	R1	RF
		μH	μF	μF	OHM/W	OHM
400	401AAG01	400	22400	44800	1000/800	8.3 4X33.2Ω IN PAR.
300	401AAG02	530	16800	33600	1500/400	11.1 3X33.2Ω IN PAR.
200	402AAG01	800	11200	22400	1500/400	16.6 2X33.2Ω IN PAR.
100	403AAG01	1600	5600	11200	2500/200	16.6 2X33.2Ω IN PAR
50	403AAG02	3200	2800	5600	2500/200	33.2Ω

TABLE II

KVA	PHASE MODULE	CONVERTER MODULE	INVERTER ASSEMBLY	COMM. P/S
400/300	36C764882AA	36C764887AA	36D868885AA	36D868889AA
200/	36C764883AA	36C764888AA	36D868886AA	36D868890AA
100/50	36C764884AA	36C764889AA	36D868887AA	36D868891AA

REV NO.	
0	
1	

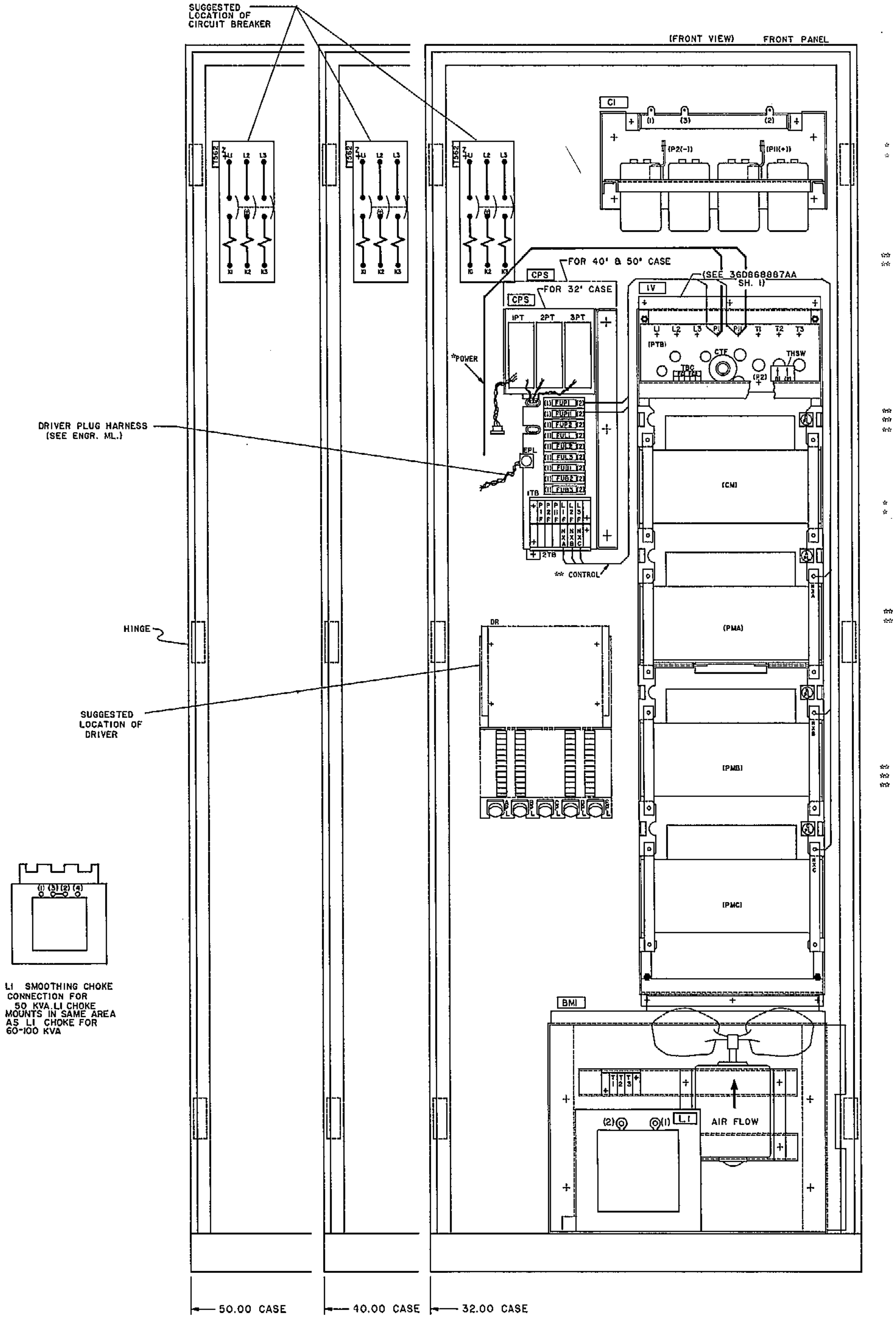
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS		
1.	May 20, 1977	HB

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M.A. Pugh AUG. 17, 1976
ISSUED D.E. Williams Jan 31, 1977
APPROVALS
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36C764880AA
CONT ON SHEET SH NO.

FF-808-F (8-78)
PRINTED IN U.S.A.


SUGGESTED LOCATION OF CIRCUIT BREAKER
(FRONT VIEW)
FRONT PANEL
C1
(1)
(3)
(2)
(P2(-))
(P11(+))
1SS2
L1 L2 L3
K1 K2 K3
CPS
FOR 40" & 50" CASE
CPS
FOR 32" CASE
(SEE 36D868887AA SH. 1)
1V
1PT
2PT
3PT
*POWER
L1 L2 L3 P1 P11 T1 T2 T3
(PTB)
TBC
CTF
THSW
(P2)
FUP1
FUP2
FUP11
FUL1
FUL2
FUL3
FUB1
FUB2
FUB3
EPL
1TB
2TB
** CONTROL
DRIVER PLUG HARNESS (SEE ENGR. ML.)
(CM)
(PMA)
(PMB)
(PMC)
HINGE
DR
SUGGESTED LOCATION OF DRIVER
(1) (3) (2) (4)
L1 SMOOTHING CHOKE CONNECTION FOR 50 KVA. L1 CHOKE MOUNTS IN SAME AREA AS L1 CHOKE FOR 60-100 KVA
BM1
T1 T2 T3
(2)
(1)
L1
AIR FLOW
50.00 CASE
40.00 CASE
32.00 CASE

WIRE TABLE

	WIRE NO.	CONNECTION
		100 KVA
		DC POWER #4 SILICONE
*	P1	1VPTB(P1), L1(1)
*	P11	1VPTB(P11), L1(2)
		DC POWER #6 SILICONE
	P11	C1(P11(+)), 1VPTB(P11)
	P2	C1(P2(-)), 1VPTB(P2)
		CONTROL #14 BLACK
**	P1	CPS-FUP1(2), 1VPTB(P1)
**	P11	CPS-FUP11(2), 1VPTB(P11)
		AC POWER #14 BLACK
	L1	1VPTB(L1), CPS-FUB1(2)
	L2	1VPTB(L2), CPS-FUB2(2)
	L3	1VPTB(L3), CPS-FUB3(2)
	K1	CPS-FUB1(1), BM1(T1)
	K2	CPS-FUB2(1), BM1(T2)
	K3	CPS-FUB3(1), BM1(T3)
		CONTROL #14 BLACK
**	NXA	1V-PMA(NXA), CPS-2TB(NXA)
**	NXB	1V-PMB(NXB), CPS-2TB(NXB)
**	NXC	1V-PMC(NXC), CPS-2TB(NXC)
		14 BLACK
	P2	CPS-FUP2(2), 1VPTB(P2)
		50KVA
		DC POWER #4 SILICONE
*	P1	1VPTB(P1), L1(1)
*	P11	1VPTB(P11), L1(4)
	2-3	L1-(2)(3) TIED
		DC POWER #6 SILICONE
	P11	C1(P11(+)), 1VPTB(P11)
	P2	C1(P2(-)), 1VPTB(P2)
		CONTROL
**	P1	CPS-FUP1(2), 1VPTB(P1)
**	P11	CPS-FUP11(2), 1VPTB(P11)
		AC POWER #14 BLACK
	L1	1VPTB(L1), CPS-FUB1(2)
	L2	1VPTB(L2), CPS-FUB2(2)
	L3	1VPTB(L3), CPS-FUB3(2)
	K1	CPS-FUB1(1), BM1(T1)
	K2	CPS-FUB2(1), BM1(T2)
	K3	CPS-FUB3(1), BM1(T3)
		CONTROL #14 BLACK
**	NXA	1V-PMA(NXA), CPS-2TB(NXA)
**	NXB	1V-PMB(NXB), CPS-2TB(NXB)
**	NXC	1V-PMC(NXC), CPS-2TB(NXC)
		#14 BLACK
	P2	CPS-FUP2(2), 1VPTB(P2)
		PLUG HARNESS WIRING TO DRIVER (FOR 50 & 100 KVA)
		RESISTOR WIRES
	P1F W	1TB(P1F), EPL(19)
	P2F R	1TB(P2F), EPL(20)
	P11F W	1TB(P11F), EPL(24)
	P2F R	1TB(P2F), EPL(25)
	L1F R	1TB(L1F), EPL(28)
	L2F W	1TB(L2F), EPL(27)
	L3F B	1TB(L3F), EPL(26)
		PLUG WIRING
	APL	DRIVER(APL), PMA(APL)
	BPL	DRIVER(BPL), PMB(BPL)
	CPL	DRIVER(CPL), PMC(CPL)
	DPL	DRIVER(DPL), CM(DPL)
	EPL	DRIVER(EPL), CPS(EPL)
	EPL22 W	DRIVER(EPL), TBC(FC1)
	EPL23 R	DRIVER(EPL), TBC(FC2)

NOTES:

1. CONNECTION DIAG. APPLIES FOR EITHER A 100 KVA OR A 50 KVA INVERTER ASM CONNECTION IN EITHER A 32", 40" OR A 50" CASE.
2. SEE ENGR'S ML FOR CASE NO., CIRCUIT BREAKER OR FUSES, DRIVER/REGULATOR, AND PLUG HARNESSES FOR APL, BPL, CPL, DPL & EPL.
3. MOUNT FUSES(WHEN USED) IN RELATIVE LOCATION OF CIRCUIT BREAKER.
4. STAMP & MOUNT LABELS(104X905BAP10 OR P11) SHOWN IN RECTANGLES ▭ .
5. FACTORY- DO NOT RUN POWER WIRING WITH INTERCONNECTION WIRE BUNDLE.
6. SEE INVERTER ASM DIAG. 36D877322AC FOR L1 LOCATION IN PLENDUM.
7. POWER WIRES (*) P1 & P11 TO BE RUN SEPARATELY FROM CONTROL WIRES(**) P1, P11, NXA, NXB, NXC AS INDICATED.

TITLE CONNECTION DIAGRAM
INVERTER 100/50 KVA
FIRST MADE FOR AF400 FCF ON 33IX403AAGI
CASE-32/40/50W X 90H 20D
36D868887AA
CONT ON SHEET FL SH NO 2

REVISIONS			PRINTS TO	
4	MAY 17, 1978 S.F. Holsopple	MAY 22 J. Bender	5B(8)M	5J(BW)
CHOKE MTG SUPPORT ADDED			5R(2)BW	
5	AUG. 18, 1978	SV-874 #29	5L(2)BW	
ADDED NOTE 5			AW(BW)	
6	MARCH 12, 1979	SV-878 #15	5D(CD)	
RELOCATED "L1" CHOKE ON BASE IN PLENUM			JA(CD)	

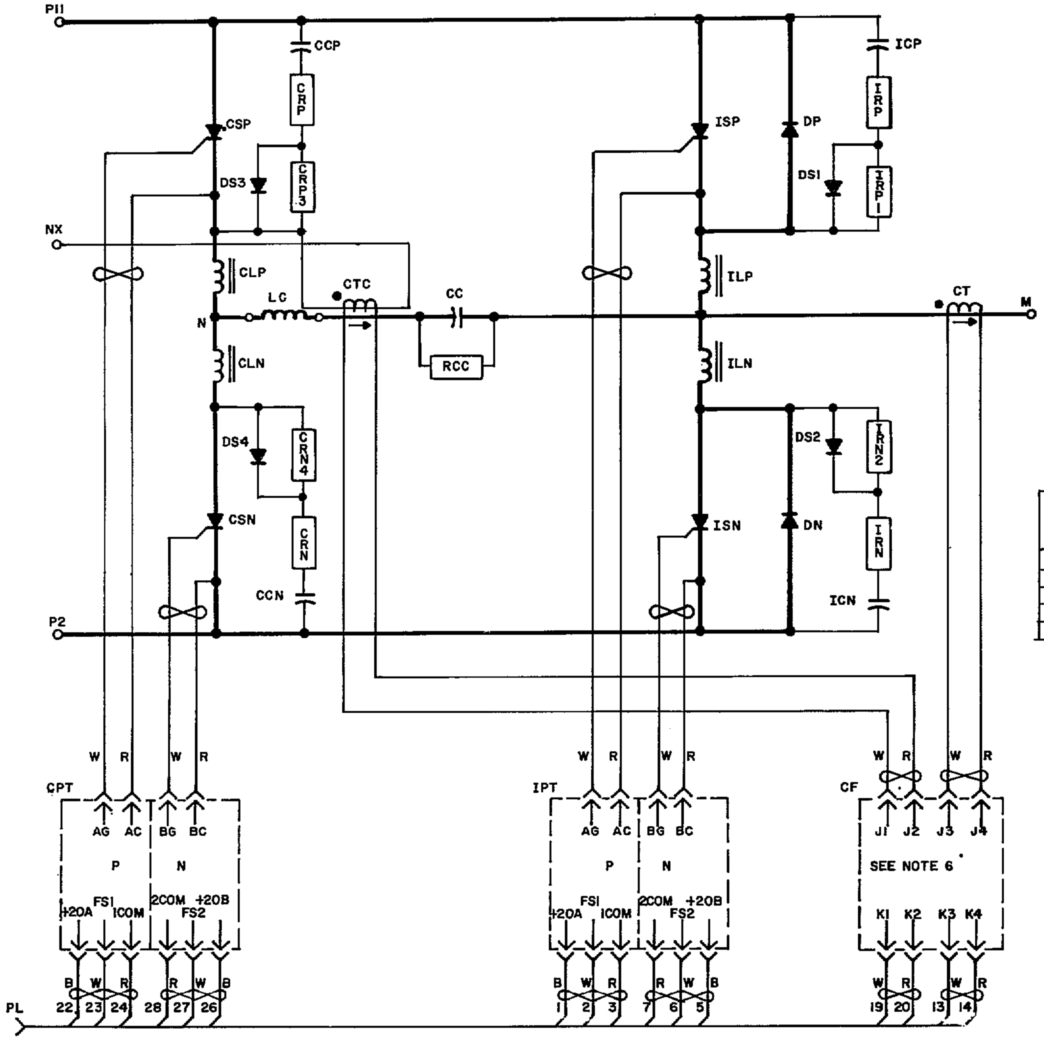
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

MADE BY R. W. GRONOSTAG JAN. 11, 1977
RE-ISSUED Feb 7, 1977
APPROVALS RW
SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION
36D868887AA
CONT ON SHEET FL SH NO 2

GENERAL ELECTRIC 36C764878AA

TITLE ELEMENTARY DIAGRAM
AF-400 PHASE MODULE
FIRST MADE FOR AF-400
331X411, 412, 413AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±



PI1
NX
P2
PL
M
CCP
CRP
CSP
DS3
CRP3
CLP
LC
N
CTC
CC
RCC
CLN
DS4
CRN4
CSN
CRN
CCN
ISP
DP
ICP
IRP
DS1
IRP1
ILP
ILN
DS2
IRN2
ISN
DN
IRN
ICN
CT
W R W R
CPT
AG AC BG BC
P N
FS1 +20A 1COM
2COM +20B FS2
B W R R W B
22 23 24 28 27 26
IPT
1 2 3 7 6 5
CF
J1 J2 J3 J4
SEE NOTE 6
K1 K2 K3 K4
W R W R
19 20 13 14

NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - DS — SNUBBER DIODE
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B — INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER.
   - 2 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - —>>— QUICK DISCONNECT "STAB-ON" CONNECTION.
   - — DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5.

KVA	331X	CC	LC	IRP, CRP IRN, CRN	IRP1, CRP3 IRN2, CRN4	ICP, CCP ICN, CCN	ILP ILN	CLP CLN	RCC
		μF	μH	OHM/W	OHM/W	μF	SAT μH	SAT μH	K OHM /W
400	411AAG01	80	5	2.5/40	20/40	1	1.5	1.5	280/4
300	411AAG02	60	7.5	2.5/40	20/40	1	2	2	280/4
200	412AAG01	40	10	2.5/40	20/40	1	2.6	2.6	560/2
100	413AAG01	20	25	10/40	20/40	.25	4	4	560/2
50	413AAG02	10	53	10/40	20/40	.25	8	8	1000/2

6. THE CURRENT FEEDBACK CARD WILL CHANGE DEPENDING ON KVA RATING. GROUP OF CARD AND JUMPER SELECTION FOR NOMINAL CURRENT ARE—

KVA	CF 193X382AA	NOMINAL INV. CURRENT A RMS	JUMPER SELECTION	NOMINAL COMM. CURRENT A PEAK
400	G05	580 500 420	XC XB XA	2100
300	G04	420 350 280	XC XB XA	1500
200	G03	280 210 175	XC XB XA	1000
100	G02	140 105 84	XC XB XA	500
50	G01	70 56 42	XC XB XA	250

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1 May 20, 1977 J. Bender

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY AUG. 16, 1976
ISSUED Jan. 31 1977
APPROVALS
SPEED VARIATOR
ERIE, PA
36C764878AA
CONT ON SHEET SH NO.

FF-608-F (9-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC | 36C 764 884AA
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED-PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

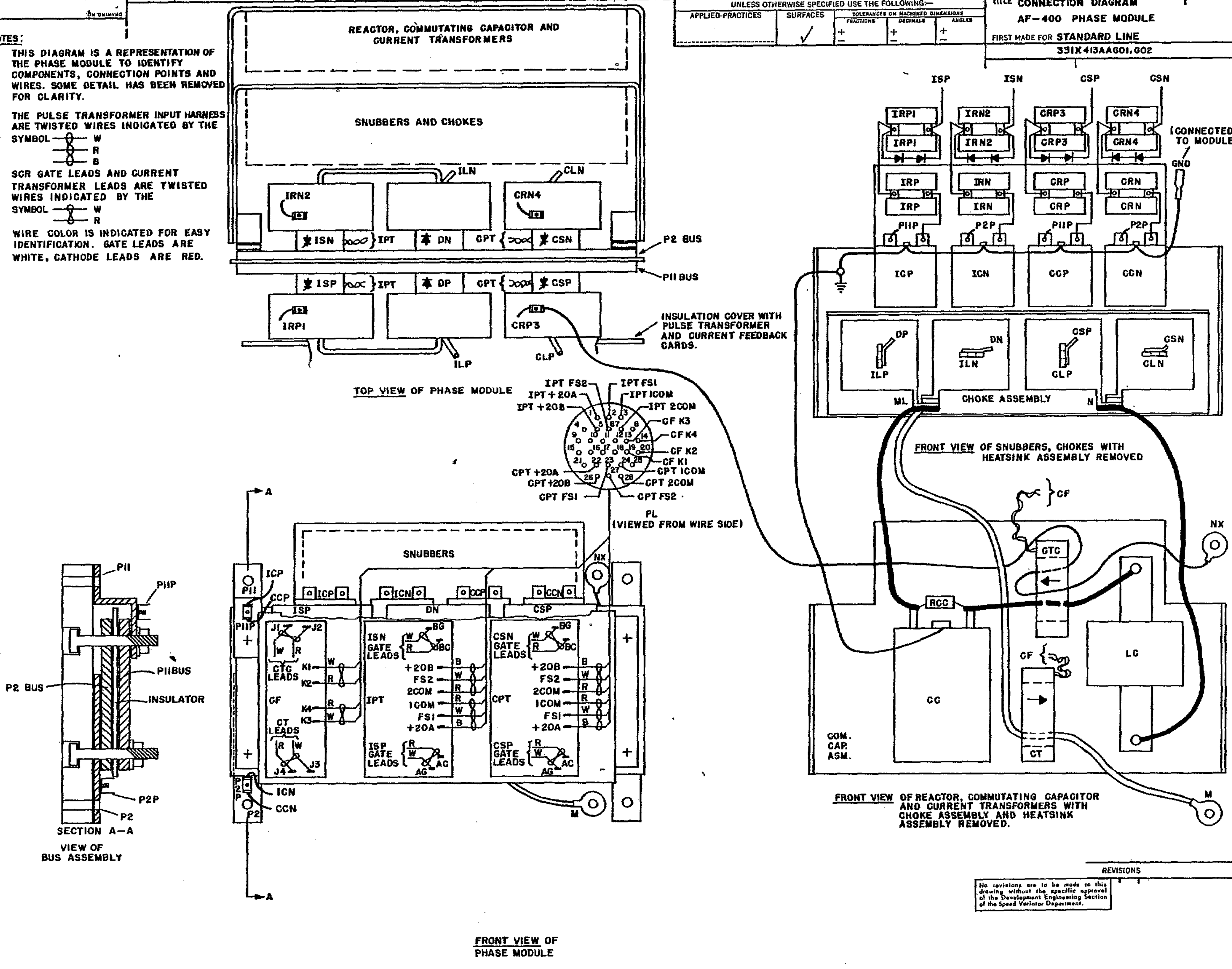
TITLE **CONNECTION DIAGRAM**
AF—400 PHASE MODULE
FIRST MADE FOR **STANDARD LINE**
331X413AAG01, G02

NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE PHASE MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.
2. THE PULSE TRANSFORMER INPUT HARNESS ARE TWISTED WIRES INDICATED BY THE SYMBOL W R B

   SCR GATE LEADS AND CURRENT TRANSFORMER LEADS ARE TWISTED WIRES INDICATED BY THE SYMBOL W R

   WIRE COLOR IS INDICATED FOR EASY IDENTIFICATION. GATE LEADS ARE WHITE, CATHODE LEADS ARE RED.


REACTOR, COMMUTATING CAPACITOR AND CURRENT TRANSFORMERS
SNUBBERS AND CHOKES
ILN
CLN
IRN2
CRN4
ISN
IPT
DN
CPT
CSN
P2 BUS
PII BUS
ISP
DP
CSP
IRPI
CRP3
ILP
CLP
INSULATION COVER WITH PULSE TRANSFORMER AND CURRENT FEEDBACK CARDS.
IPT FS2
IPT FSI
IPT +20A
IPT ICOM
IPT +20B
IPT 2COM
CF K3
CF K4
CF K2
CF KI
CPT +20A
CPT ICOM
CPT +20B
CPT 2COM
CPT FSI
CPT FS2
PL
(VIEWED FROM WIRE SIDE)
SNUBBERS
ICP
CCP
ICN
CCN
NX
M
PII
PIIP
P2P
P2
J1
J2
J3
J4
W
R
B
CTC LEADS
CT LEADS
K1
K2
K3
K4
GF
ISN GATE LEADS
ISP GATE LEADS
CSN GATE LEADS
CSP GATE LEADS
BG
BC
AG
AC
+20B
FS2
2COM
1COM
FS1
+20A
A
PIIBUS
INSULATOR
ISP
ISN
CSP
CSN
IRPI
IRN2
CRP3
CRN4
IRP
IRN
CRP
CRN
(CONNECTED EXTERNAL TO MODULE)
GND
IGP
ICN
CGP
CGN
CHOKE ASSEMBLY
ML
N
CF
GTC
RCC
CC
LC
CT
COM. CAP. ASM.

TOP VIEW OF PHASE MODULE

FRONT VIEW OF PHASE MODULE

SECTION A—A
VIEW OF BUS ASSEMBLY

FRONT VIEW OF SNUBBERS, CHOKES WITH HEATSINK ASSEMBLY REMOVED

FRONT VIEW OF REACTOR, COMMUTATING CAPACITOR AND CURRENT TRANSFORMERS WITH CHOKE ASSEMBLY AND HEATSINK ASSEMBLY REMOVED.

REV NO. 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(3)23	JA(CD)
5D(BK)	
5(QC)B	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY E. Quattrini DEC. 20, 1976
ISSUED D. LaRose DEC.28,1976
APPROVALS
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36C 764 884AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

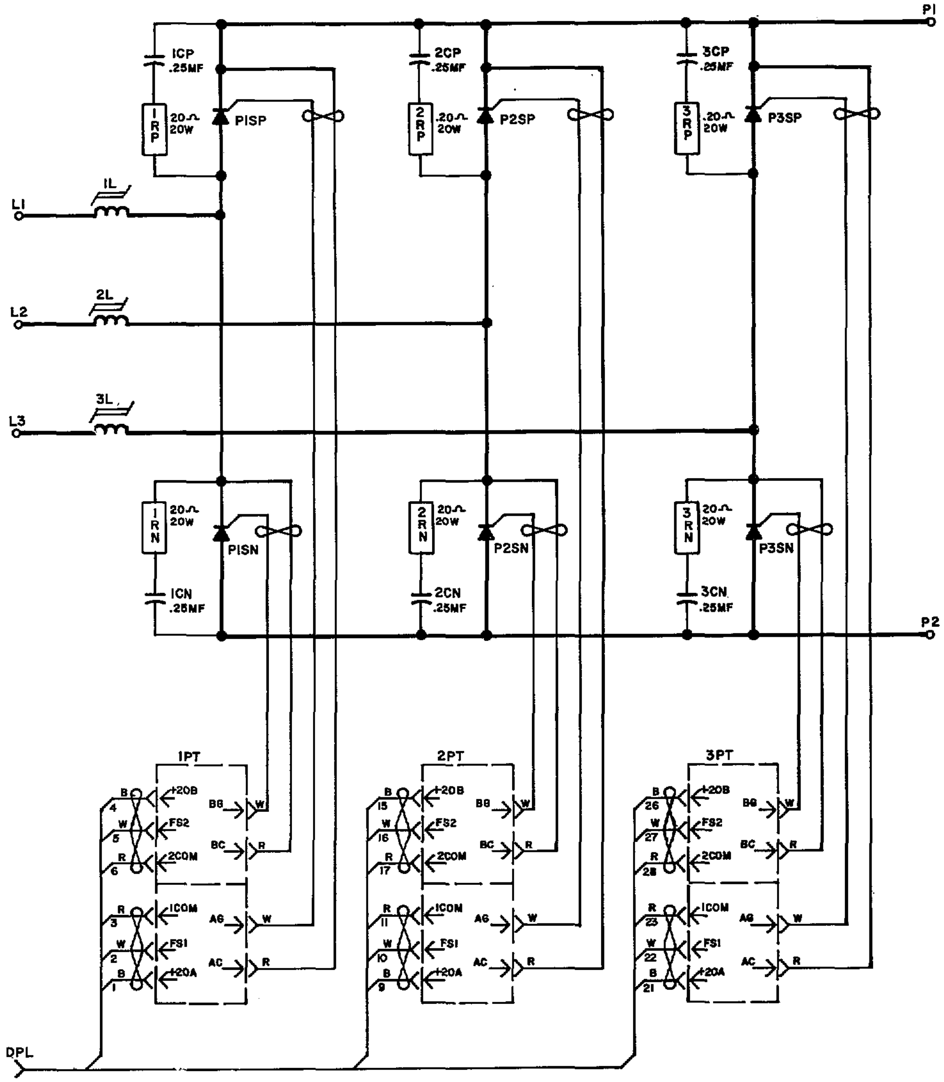
FP-80B-F (4-72)
PRINTED IN U.S.A.

GENERAL ELECTRIC 36C764879AA

TITLE	ELEMENTARY DIAGRAM AF-400 CONVERTER MODULE
FIRST MADE FOR	STANDARD LINE 331X421, 422, 423AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	√	±	±	±



P1
P2
L1
L2
L3
1L
2L
3L
1CP .25MF
2CP .25MF
3CP .25MF
1RP 20Ω 20W
2RP 20Ω 20W
3RP 20Ω 20W
P1SP
P2SP
P3SP
1RN 20Ω 20W
2RN 20Ω 20W
3RN 20Ω 20W
P1SN
P2SN
P3SN
1CN .25MF
2CN .25MF
3CN .25MF
1PT
2PT
3PT
+20B
FS2
2COM
1COM
FS1
+20A
BG
BC
AG
AC
W
R
B
1
2
3
4
5
6
9
10
11
15
16
17
21
22
23
26
27
28
DPL

NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES P1, P2 ETC., TO INDICATE PHASE CONTROL AND PHASE. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - X — AUXILIARY SCR
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B — INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER
   - 1 2 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.
   - →≻— — QUICK DISCONNECT "STAB ON" CONNECTION
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5. 1L, 2L, 3L; 3 FERRITE CORES 1 TURN

REV NO.
0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

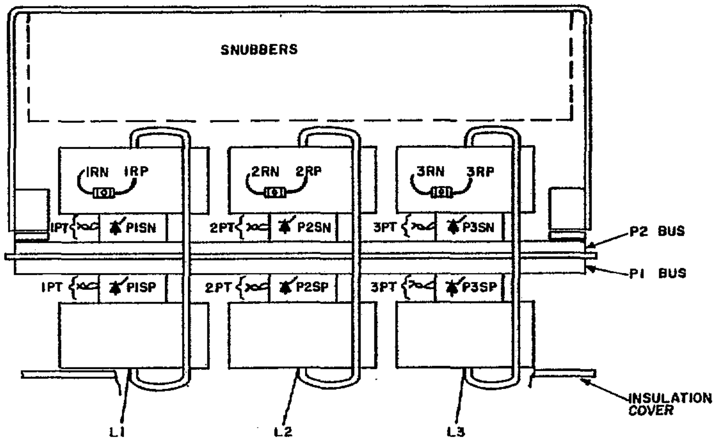
MADE BY M D Pazzola	AUG. 24, 1976	APPROVALS EAS	SPEED VARIATOR	36C764879AA
ISSUED D E Williams	[illegible] 31 1977		ERIE, PA	

36C 764889AA

GENERAL ELECTRIC | 36C 764889AA CONT ON SHEET SH NO.

TITLE CONNECTION DIAGRAM
AF-400 CONVERTER MODULE
FIRST MADE FOR STANDARD LINE
331X423AAG01

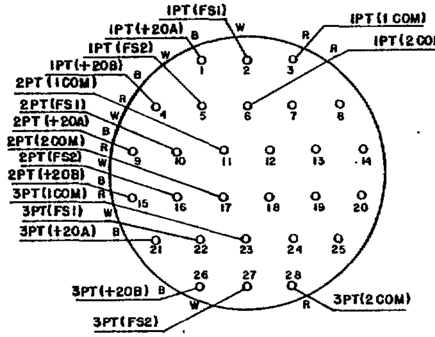
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±


SNUBBERS
1RN 1RP
2RN 2RP
3RN 3RP
1PT P1SN
2PT P2SN
3PT P3SN
1PT P1SP
2PT P2SP
3PT P3SP
P2 BUS
P1 BUS
INSULATION COVER
L1
L2
L3

TOP VIEW OF CONVERTER MODULE

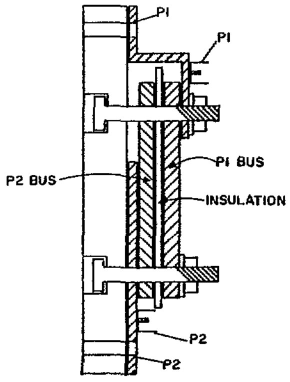
NOTES:

1. THIS DIAGRAM IS A REPRESENTATION OF THE CONVERTER MODULE TO IDENTIFY COMPONENTS, CONNECTION POINTS AND WIRES. SOME DETAIL HAS BEEN REMOVED FOR CLARITY.

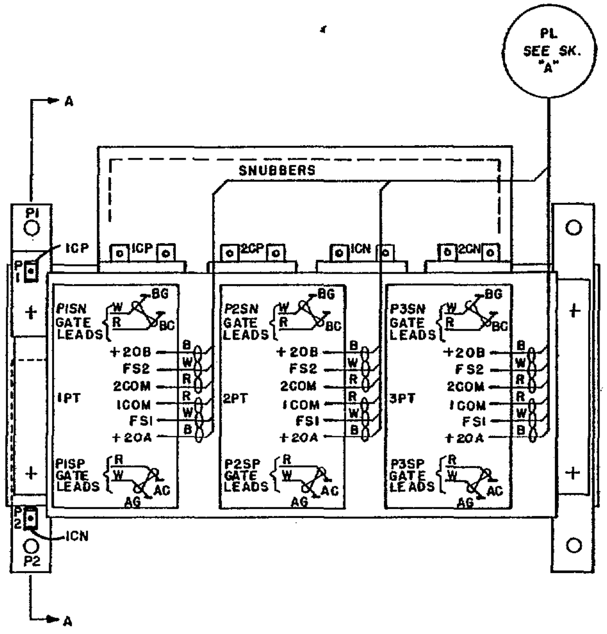

1PT(FS1) W
1PT(+20A) B
1PT(FS2) W
1PT(+20B) B
2PT(1COM) R
2PT(FS1) W
2PT(+20A) B
2PT(2COM) R
2PT(FS2) W
2PT(+20B) B
3PT(1COM) R
3PT(FS1) W
3PT(+20A) B
3PT(+20B) B
3PT(FS2) W
3PT(2COM) R
1PT(1 COM) R
1PT(2 COM) R
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28

SKETCH "A"

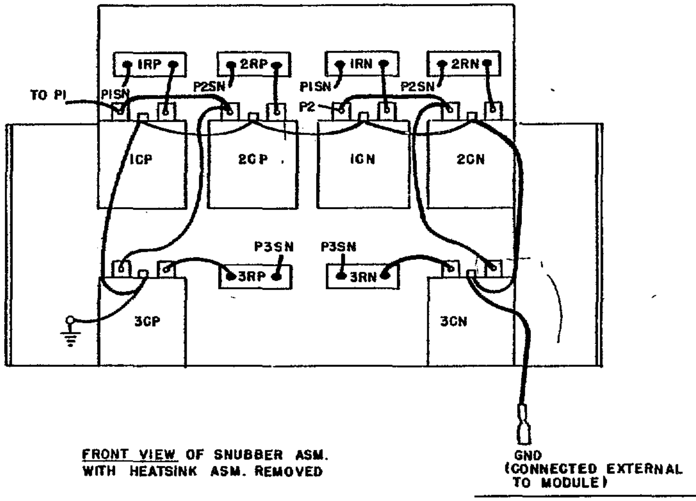
VIEW OF WIRED END OF PL (FREE END)


P1
P1
P1 BUS
INSULATION
P2 BUS
P2
P2

SECTION A–A
VIEW OF BUS ASSEMBLY


PL SEE SK. "A"
A
A
SNUBBERS
1CP
1CP
2CP
1CN
2CN
P1
P2
1CN
P1SN GATE LEADS
P2SN GATE LEADS
P3SN GATE LEADS
W R BG BC
+20B
FS2
2COM
1COM
FS1
+20A
B W R R W B
1PT
2PT
3PT
P1SP GATE LEADS
P2SP GATE LEADS
P3SP GATE LEADS
R W AG AC
+
+
+
+

FRONT VIEW OF CONVERTER MODULE


1RP
2RP
1RN
2RN
TO P1
P1SN
P2SN
P1SN
P2SN
P2
1CP
2CP
1CN
2CN
P3SN
P3SN
3RP
3RN
3CP
3CN
GND
(CONNECTED EXTERNAL TO MODULE)

FRONT VIEW OF SNUBBER ASM.
WITH HEATSINK ASM. REMOVED

36C 764889 AA
REV NO. 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(3)23	JA(CD)
5D(BK)	
5(QC)8	
5QC(BK)5	
5(TS)50	
5(S)	

MADE BY E. Gualtier JAN. 6, 1977
ISSUED D. Forbes JAN 7, 1977
APPROVALS RJV SAS
SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION
36C 764 889AA CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

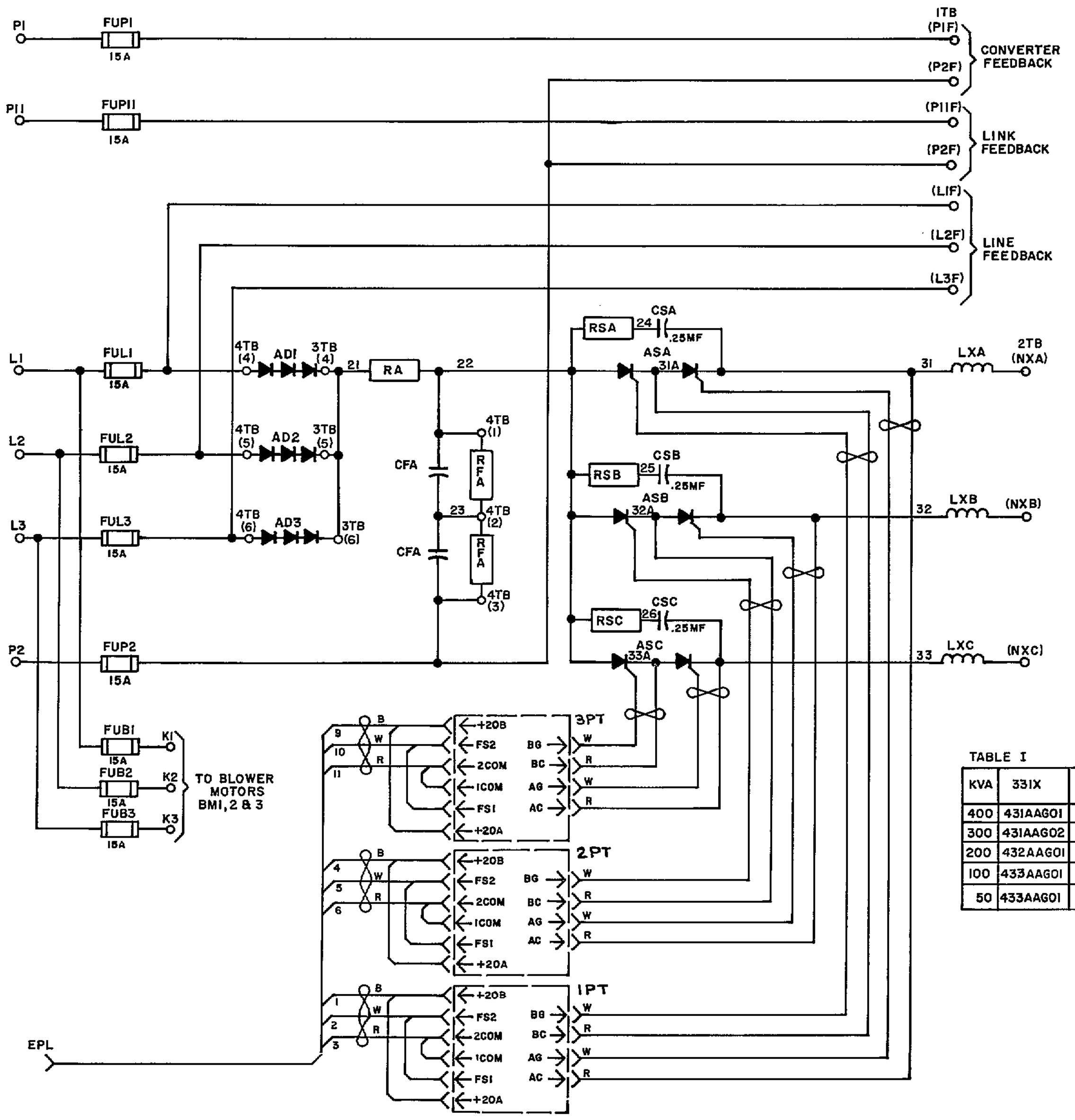
GENERAL ELECTRIC

36C764881AA
CONT ON SHEET SH NO.

TITLE ELEMENTARY DIAGRAM
COMMUTATION POWER SUPPLY
FIRST MADE FOR AF-400
331X431AAG01

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±



P1
FUP1
15A
ITB (P1F)
(P2F)
CONVERTER FEEDBACK
P11
FUP11
15A
(P11F)
(P2F)
LINK FEEDBACK
(L1F)
(L2F)
(L3F)
LINE FEEDBACK
L1
FUL1
15A
L2
FUL2
15A
L3
FUL3
15A
P2
FUP2
15A
4TB (4)
AD1
3TB (4)
4TB (5)
AD2
3TB (5)
4TB (6)
AD3
3TB (6)
21
RA
22
4TB (1)
RFA
23
4TB (2)
RFA
4TB (3)
CFA
CFA
RSA
24
CSA
.25MF
ASA
31A
RSB
25
CSB
.25MF
ASB
32A
RSC
26
CSC
.25MF
ASC
33A
31
LXA
2TB (NXA)
32
LXB
(NXB)
33
LXC
(NXC)
FUB1
15A
FUB2
15A
FUB3
15A
K1
K2
K3
TO BLOWER MOTORS BM1, 2 & 3
3PT
2PT
1PT
+20B
FS2
2COM
1COM
FS1
+20A
BG
BC
AG
AC
B
W
R
9
10
11
4
5
6
1
2
3
EPL

TABLE I

KVA	331X	LX µH	CFA µF	RA OHMS/W	RFA OHM/CONN/W	RSA OHM/CONN/W
400	431AAG01	54	1400	5/100	10K/2X5K SER/20	100/2 X 200 PAR/40
300	431AAG02	54	1400	5/100	"	"
200	432AAG01	110	850	10/50	"	200/2X100 SER/40
100	433AAG01	220	850	10/50	"	400/2X200 SER/40
50	433AAG01	"	850	10/50	"	"

REVISIONS

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1	Feb 1, 1977 E. Quattrin DW	SV872 #12A
2	NOV. 7, 1977 G.R. Baker	PER #1B SV-872 #13A
3	JAN 31, 1978 [illegible]	SV872 #20
4	M. Gotty June 21, 78	PER [illegible] SV-872 #8

CHG MARKING
1PT WAS PTA
2PT WAS PTB
3PT WAS PTC

PRINTS TO

5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY M.B. Pazka AUG 25, 1976
ISSUED W. Kazanauskas OCT 18, 1976
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764881AA
CONT ON SHEET SH NO.

REV NO. 0 1 2 3 4

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWING NO.
CONT ON SHEET SH NO.

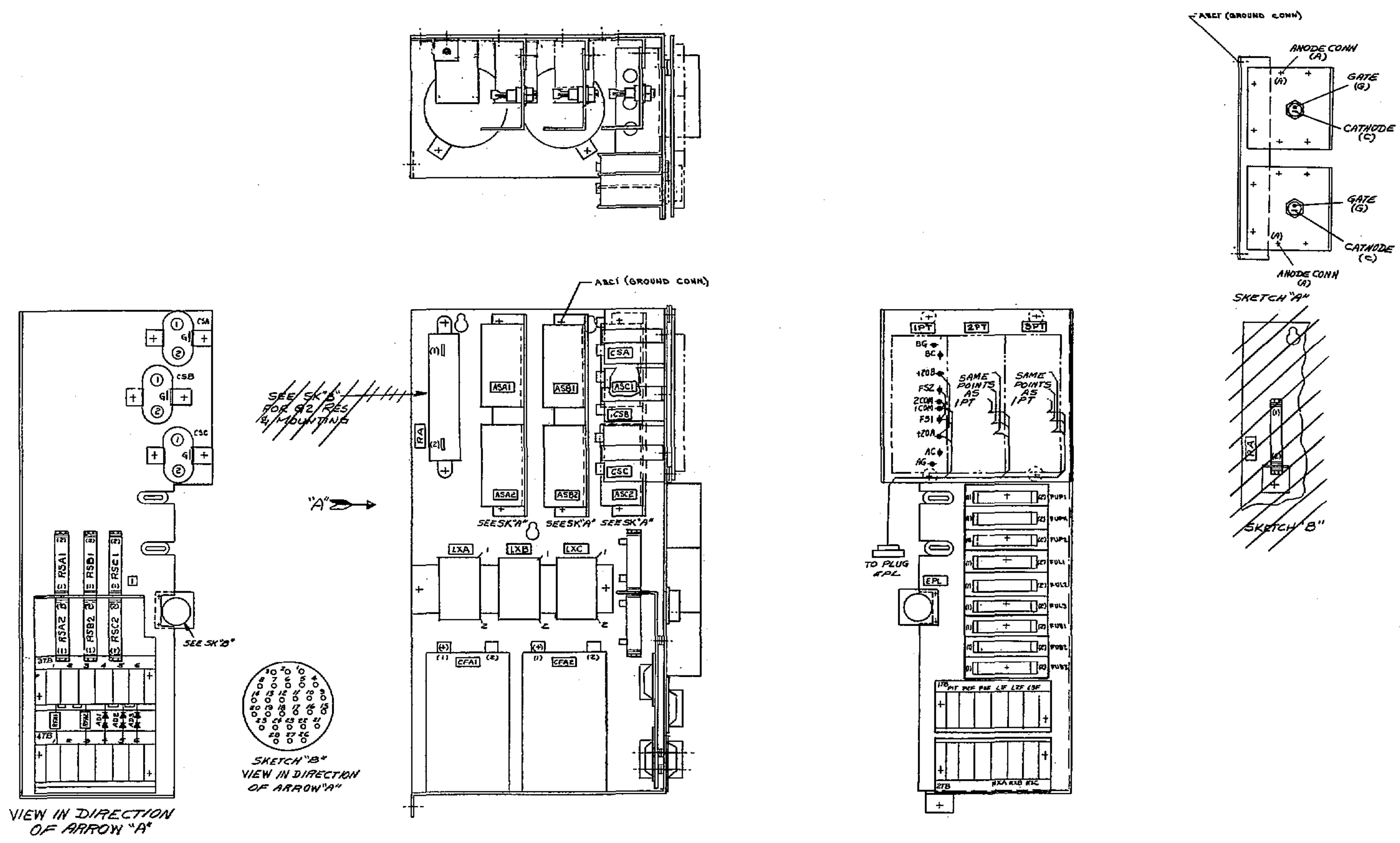
FF-809-F (9-75) PRINTED IN U.S.A.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES•	SURFACES	FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

GENERAL ELECTRIC 36D868891AA

TITLE CONNECTION DIAGRAM
COMMUTATION POWER SUPPLY
FIRST MADE FOR AF400
100/50KVA
331X433AAG1~~/G2~~



ASC1 (GROUND CONN)
ANODE CONN (A)
GATE (G)
CATHODE (C)
(A)
GATE (G)
CATHODE (C)
ANODE CONN (A)
SKETCH "A"
RA
SKETCH "B"
ASC1 (GROUND CONN)
CSA
CSB
CSC
ASC1
ASA1
ASB1
ASA2
ASB2
ASC2
RA
SEE SK"B" FOR R2 RES. & MOUNTING
"A"
SEE SK"A"
SEE SK"A"
SEE SK"A"
LXA
LXB
LXC
CFA1
CFA2
RSA1
RSB1
RSC1
RSA2
RSB2
RSC2
3TB
4TB
SEE SK"B"
SKETCH "B" VIEW IN DIRECTION OF ARROW "A"
VIEW IN DIRECTION OF ARROW "A"
1PT
2PT
3PT
BG
BC
+20B
FS2
2COM
1COM
FS1
+20A
AC
AG
SAME POINTS AS 1PT
SAME POINTS AS 1PT
FUP1
FUP2
FUP11
FUL1
FUL2
FUL3
FUB1
FUB2
FUB3
EPL
TO PLUG EPL
1TB
P1F P2F P11F L1F L2F L3F
2TB
NXA NXB NXC

WIRE NO.	CONNECTION
	#16 BLACK - POWER WIRING
P1F	FUP1(1), 1TB(P1F)
P2F	1TB(P2F), FUP2(1), CFA2(2), 4TB(3)
P11F	1TB(P11F), FUP11(1)
L1F	1TB(L1F), FUL1(1), 4TB(4)
L2F	1TB(L2F), FUL2(1), 4TB(5)
L3F	1TB(L3F), FUL3(1), 4TB(6)
21	3TB(4)*, 3TB(5)*, 3TB(6)*, RA(2)
22	4TB(1), CFA1(1), ASC1(A), ASB1(A), ASA1(A), RA(1)
~~23~~	~~3TB(1)*, 3TB(2)*, 3TB(3)*, CFA2(1), CFA1(2) FOR 50KVA ONLY~~
22	ASA1(A), RSC1(2), RSA1(2), RSB1(2)
24	CSA(1), RSA2(2)
24A	RSA1(1), RSA2(1)
25	CSB(1), RSB2(2)
25A	RSB1(1), RSB2(1)
26	CSC(1), RSC2(2)
26A	RSC1(1), RSC2(1)
31	CSA(2), LXA(1), ASA2(C)▲
32	CSB(2), LXB(1), ASB2(C)▲
33	CSC(2), LXC(1), ASC2(C)▲
NXA	2TB(NXA), LXA(2)
NXB	2TB(NXB), LXB(2)
NXC	2TB(NXC), LXC(2)
31A	ASA1(C)▲ ASA2(A)
32A	ASB1(C)▲ ASB2(A)
33A	ASC1(C)▲ ASC2(A)
G	CSC(G), CSB(G), CSA(G), ASC1(G)
L1	FUL1(2), FUB1(2)
L2	FUL2(2), FUB2(2)
L3	FUL3(2), FUB3(2)
23	4TB(2), CFA2(1), CFA1(2) ~~FOR G1=100KVA ONLY~~
	*METAL JUMPER CONNECTION
	#20 RED & WHITE TWISTED-SCR GATE LEAD WIRING
W	ASA1(G), 1PT(BG)
R	ASA1(C)▲ 1PT(BC)
W	ASA2(G), 1PT(AG)
R	ASA2(C)▲ 1PT(AC)
W	ASB1(G), 2PT(BG)
R	ASB1(C)▲ 2PT(BC)
W	ASB2(G), 2PT(AG)
R	ASB2(C)▲ 2PT(AC)
W	ASC1(G), 3PT(BG)
R	ASC1(C)▲ 3PT(BC)
W	ASC2(G), 3PT(AG)
R	ASC2(C)▲ 3PT(AC)
	#20 BLUE, WHITE & RED TWISTED - PLUG WIRING (EPL)
B	EPL(1), 1PT(+20B), 1PT(+20A)**
W	EPL(2), 1PT(FS2), 1PT(FS1)**
R	EPL(3), 1PT(2COM), 1PT(1COM)**
B	EPL(4), 2PT(+20B), 2PT(+20A)**
W	EPL(5), 2PT(FS2), 2PT(FS1)**
R	EPL(6), 2PT(2COM), 2PT(1COM)**
B	EPL(9), 3PT(+20B), 3PT(+20A)**
W	EPL(10), 3PT(FS2), 3PT(FS1)**
R	EPL(11), 3PT(2COM), 3PT(1COM)**
	** LAST CONNECTION POINT NOT TWISTED
	▲ SOLDER BARE END OF THE #16 BLACK WIRE WITH THE BARE END OF THE #20 RED WIRE TO THE CATHODE (C) OF THE SCR TOGETHER AS ONE OPERATION.

REVISIONS

4 SEPT 19, 1978	WHITE SCR GATE LEAD WAS ASA2(C), 1PT(AG)
3 JAN 31, 1978 #20	CHG LABEL MARKING ON ASM 1PT WAS PTA 2PT WAS PTB, 3PT WAS PTC. GATE LEAD WIRES WERE PTA, PTB, PTC TWISTED PLUG WIRING WAS PTA, PTB, & PTC
2	[illegible]
1 MAY 19, 1976	ADDED SK"B", 100KVA, G2. NOTED WIRE NO. 23 FOR 50KVA (G2) AND ADDED WIRE NO 23 FOR 100KVA (G1) TO WIRE TABLE.

PRINTS TO

SB(CM)	SJ(BW)
BR(EB)	
BLT(EB)	
AW(BW)	
SX(CO)	
JA(CO)	

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

Sept 27, 1976
Oct 18, 1976

SPEED VARIATOR
ERIE, PA.

36D868891AA